AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

FINAL ASSESSMENT TEST

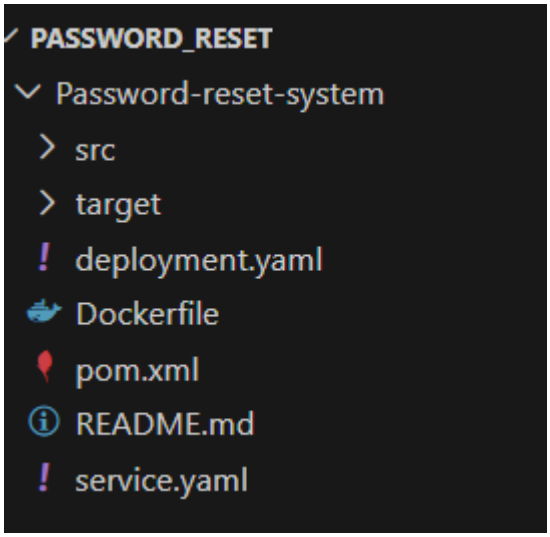
COURSE CODE :ISWE406P

FACULTY:DR.N.BALAGANESH

NAME:GOPIKA.S

REGISTER NUMBER:23MIS0560

SET-5



pom.xml | Dockerfile | deployment.yaml | service.yaml

Password-reset-system > pom.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>simple-maven-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

</project>
```

Password-reset-system > src > main > java > com > example > App.java

```java
package com.example;

import java.util.Random;

public class App {

    private String generatedOTP;

    // generate OTP
    public String generateOTP() {
        Random random = new Random();
        generatedOTP = String.valueOf(1000 + random.nextInt(9000));
        return generatedOTP;
    }

    // verify OTP
    public boolean verifyOTP(String inputOTP) {
        return generatedOTP != null && generatedOTP.equals(inputOTP);
    }

    // reset password
    public String resetPassword(String otp, String newPassword) {
        if (!verifyOTP(otp)) {
            throw new IllegalArgumentException("Invalid OTP attempt");
        }

        if (newPassword.length() < 6) {
            throw new IllegalArgumentException("Weak password");
        }

        return "Password Reset Successful";
    }

```

Password-reset-system > src > test > java > com > example > AppTest.java

```java
package com.example;

import org.junit.Test;
import static org.junit.Assert.*;

public class AppTest {

    App system = new App();

    @Test(expected = IllegalArgumentException.class)
    public void testInvalidOTP() {
        system.generateOTP();
        system.resetPassword("0000", "newpass123");
    }

    @Test(expected = IllegalArgumentException.class)
    public void testWeakPassword() {
        String otp = system.generateOTP();
        system.resetPassword(otp, "123");
    }

    @Test
    public void testValidResetFlow() {
        String otp = system.generateOTP();
        String result = system.resetPassword(otp, "securePass");
        assertEquals("Password Reset Successful", result);
    }
}
```

Password-reset-system > ! deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: auth-deployment

spec:
  replicas: 2

  selector:
    matchLabels:
      app: auth

  template:
    metadata:
      labels:
        app: auth

    spec:
      containers:
      - name: auth-container
        image: yourdockerhub/auth-system:latest
        ports:
        - containerPort: 8080
```

Password-reset-system > Dockerfile

```
FROM eclipse-temurin:17-jdk

WORKDIR /app

COPY target/*.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

```
PS C:\Users\gopik\OneDrive\Desktop\password_reset> git init
Initialized empty Git repository in C:/Users/gopik/OneDrive/Desktop/password_reset/.git/
PS C:\Users\gopik\OneDrive\Desktop\password_reset> git clone https://github.com/gopika-sathish/Password-reset-system.git
Cloning into 'Password-reset-system'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (22/22), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 22 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (22/22), 4.91 KiB | 279.00 KiB/s, done.
Resolving deltas: 100% (2/2), done.
PS C:\Users\gopik\OneDrive\Desktop\password_reset> mvn clean test
```

```
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ simple-maven-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ simple-maven-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ simple-maven-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ simple-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.065 s -- in com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.984 s
[INFO] Finished at: 2026-04-13T10:17:00+05:30
[INFO] ------------------------------------------------------------------------
```

Script ?

```
pipeline {
    agent any
    environment{
        DOCKER_IMAGE="gopikasathish/password"
    }

    tools {
        maven 'M3'
    }

    stages {

        stage('Checkout Git') {
            steps {
                git branch: 'main',
```

☑ Use Groovy Sandbox ?

```
Started by user Gopikasathish
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\PASSWORD
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout Git)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gopika-sathish/Password-reset-system.git
 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\PASSWORD # timeout=10
Fetching upstream changes from https://github.com/gopika-sathish/Password-reset-system.git
 > git.exe --version # timeout=10
```

```
Started by user Gopikasathish
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\PASSWORD
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout Git)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gopika-sathish/Password-reset-system.git
 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\PASSWORD # timeout=10
Fetching upstream changes from https://github.com/gopika-sathish/Password-reset-system.git
 > git.exe --version # timeout=10
```

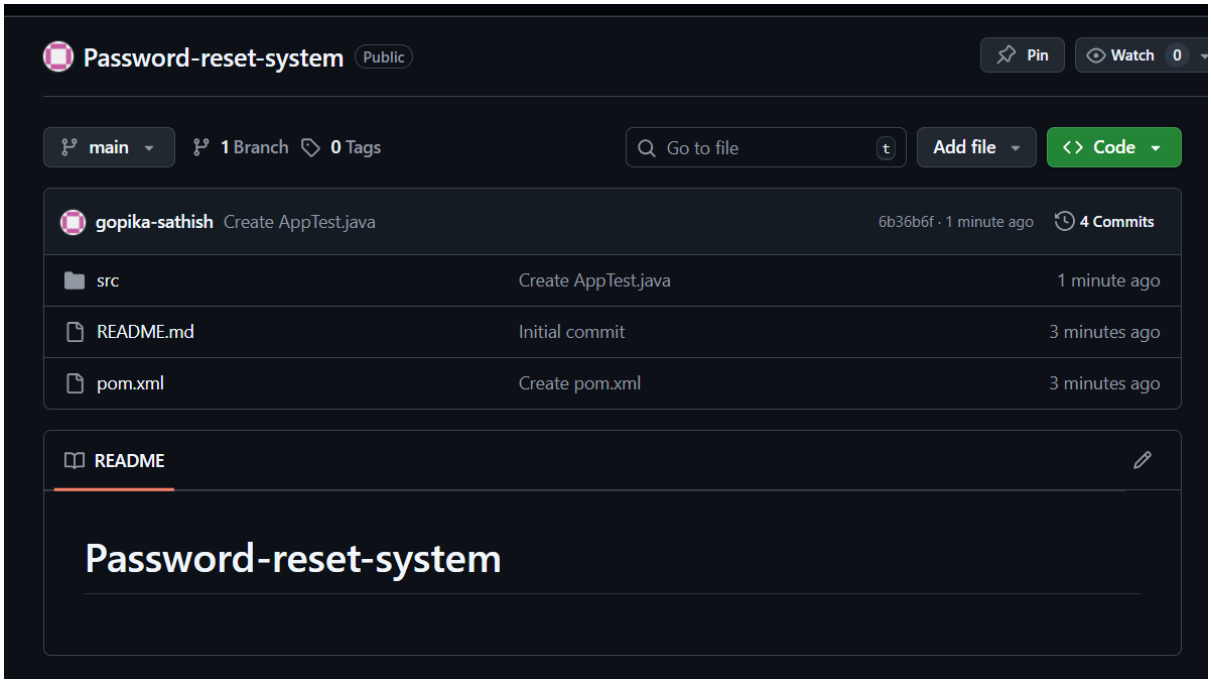
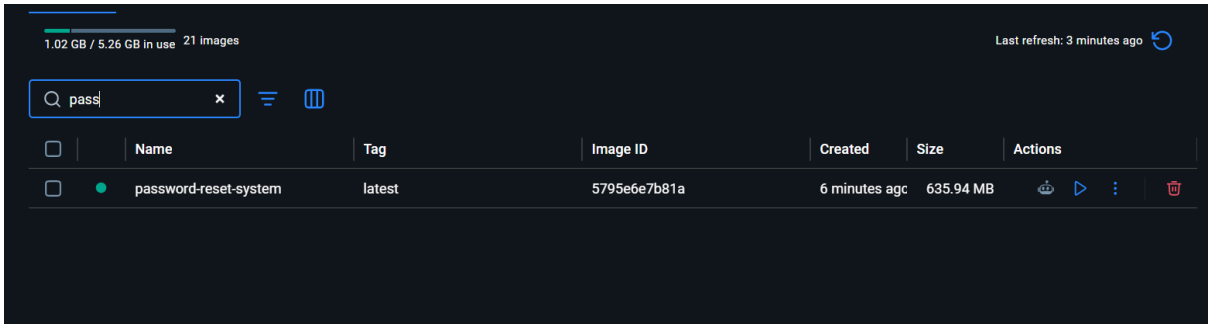
```
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> mvn clean package

[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:simple-maven-app >-------------------
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ simple-maven-app ---
[INFO] Deleting C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ simple-maven-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ simple-maven-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ simple-maven-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
```

```
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ simple-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.065 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
------------------------------------------------
[INFO] Total time:  2.974 s
[INFO] Finished at: 2026-04-13T10:28:45+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system>
```

```
S C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> dir target


   Directory: C:\Users\gopik
   \OneDrive\Desktop\passwor
   d_reset\Password-reset-sy
   stem\target


ode            LastWriteTime
---            -------------
-----    13-04-2026     10:57
-----    13-04-2026     10:57
-----    13-04-2026     10:57
-----    13-04-2026     10:57
-----    13-04-2026     10:57
-----    13-04-2026     10:57
-----    13-04-2026     10:57
a----    13-04-2026     10:57


S C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> docker build -t password-reset-system .
+] Building 4.2s (8/8) FINISH
=> [internal] load bui  0.1s
```

```
PS C:\Users\gopik\OneDrive\Desktop\passwordddocker build -t password-reset-system .
[+] Building 2.4s (8/8) FINISH
 => [internal] load bui  0.0s
 => => transferrin 144B  0.0s
 => [internal] load met  1.7s
 => [internal] load .do  0.0s
 => => transferring  2B  0.0s
 => [1/3] FROM docker.i  0.1s
 => => resolve docker.i  0.0s
 => [internal] load bui  0.1s
 => => transferr 2.85kB  0.1s
 => CACHED [2/3] WORKDI  0.0s
 => [3/3] COPY target/*  0.0s
 => exporting to image   0.3s
 => => exporting layers  0.1s
 => => exporting manife  0.0s
 => => exporting config  0.0s
 => => exporting attest  0.0s
 => => exporting manife  0.0s
 => => naming to docker  0.0s
 => => unpacking to doc  0.0s
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> docker images
          i Info →   U  In Use
IMAGE              ID
crm-app:latest     3c3d2d211643
docker/desktop-cloud-provider-kind:v0.5.0
                   4ad59ce20658
docker/desktop-containerd-registry-mirror:v0.0.3
                   179a9ea3ae47
docker/desktop-kubernetes:kubernetes-v1.34.1-cni-v1.7.1-critools-v1.33.0-cri-dockerd-v0.3.20-1-debian
                   12d6673564e0
```

```
no main manifest attribute, in app.jar
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
0:8080 password-reset-system
no main manifest attribute, in app.jar
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> kubectl apply -f deployment.yaml
>> kubectl apply -f service.yaml
deployment.apps/auth-deployment created
service/auth-service created
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system>
```

```
0:8080 password-reset-system
no main manifest attribute, in app.jar
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> kubectl apply -f deployment.yaml
>> kubectl apply -f service.yaml
deployment.apps/auth-deployment created
service/auth-service created
PS C:\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system>
```

```
in manifest attribute, in app.jar
\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> docker ps
INER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> kubectl apply -f deployment.yaml
in manifest attribute, in app.jar
\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> docker ps
INER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> kubectl apply -f deployment.yaml
bectl apply -f service.yaml
INER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> kubectl apply -f deployment.yaml
bectl apply -f service.yaml
yment.apps/auth-deployment created
\Users\gopik\OneDrive\Desktop\password_reset\Password-reset-system> kubectl apply -f deployment.yaml
bectl apply -f service.yaml
yment.apps/auth-deployment created
yment.apps/auth-deployment created
ce/auth-service created
```